Lecture 24 — Inverses, rank, and solvability

Wed Nov 4

Last updated: November 4, 2026.

Boyd video (optional): Lecture 30 — left & right inverses · Lecture 15 — linear independence (rank ideas) · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Strang / MIT (optional): Reading: Strang §3.5 (PDF) — rank / column space / nullspace
Video: 18.06 Lec 6 — column space and nullspace · MIT 18.06 playlist (Strang)

Reading: Boyd 11.1–11.3; class notes + selected Strang §3.5 (rank, $C(A)$, $N(A)$). Related objectives: Lectures. Schedule.

Monday continued Lab 3 (including the $[A \mid I]$ peek). Today names that right block A^{-1}, then connects invertibility to the RREF language you already use: column space, null space, and rank. An inverse is the **square** case where every b has exactly one x. Not a four-subspaces theory day.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Explain why reducing $[A \mid I]$ to $[I \mid B]$ gives $B = A^{-1}$, and use an inverse to solve $Ax = b$.
2. Interpret $C(A)$ as the possible outputs Ax and $N(A)$ as the inputs sent to zero.
3. Find rank from pivots and relate it to free variables and $\dim N(A)$.
4. Explain when $Ax = b$ is solvable and when a solution, if it exists, is unique.
5. Distinguish square invertibility from full column rank for a tall matrix.

Warm-up

Recall $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $E = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$.

(a) What row operation does multiplying by E perform?

(b) Compute EA.

(c) What does this tell us about solving $Ax = b$?

Fill in

Inverse from elementary matrices

If elementary matrices $E_1, \dots, E_k$ reduce a square matrix A to I, then

$$E_k \cdots E_1 A = I, \qquad A^{-1} = E_k \cdots E_1.$$

Row-reducing $[A \mid I]$ to $[I \mid B]$ therefore gives $B =$ _____. The same matrix solves every right-hand side: if $Ax = b$, then $x =$ _____.

Check. After you find a candidate A^{-1}, multiply:

$$A^{-1}A = _____ \qquad \text{(and/or } AA^{-1} = I\text{)}.$$

If the product is not I, you made an arithmetic error — or A was not invertible.

Column space and null space

The **column space** $C(A)$ is the span of the columns of A — every vector of the form Ax. So

$$Ax = b \text{ is solvable} \quad \iff \quad b \in _____.$$

The **null space** is $N(A) = \{x : Ax = ____\}$. If $Ax_p = b$, every other solution has the form _____. A solution, when one exists, is unique exactly when _____.

Rank from pivots

A **pivot** in $\text{rref}(A)$ is the leading 1 in a nonzero row. Pivot columns of A are the corresponding columns of the original matrix. Example:

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

has pivots in columns _____; column ____ is free.

Basis / dimension (recall). A basis is a linearly independent spanning set; dimension is how many vectors sit in any basis.

Rank. $\text{rank}(A) = \dim C(A) =$ the _____ of A. For an $m \times n$ matrix,

$$\dim N(A) = n - \text{rank}(A) \qquad \text{(rank–nullity)}.$$

Square case (Monday's inverse, restated)

For $n \times n$ matrices,

$$A \text{ invertible} \iff \text{rank}(A) = n \iff C(A) = \mathbb{R}^n \iff N(A) = \{0\}.$$

Rectangular matrices never have a two-sided inverse; we still use $C(A)$, $N(A)$, and rank. Later, least squares asks for the closest Ax when $b \notin C(A)$.

Worked in class

Three 2×2 inverses

W1. Find A^{-1} by reducing $[A \mid I]$. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

(a) Form $[A \mid I]$ and row-reduce to $[I \mid A^{-1}]$.
(b) **Check:** compute $A^{-1}A$ and confirm you get I.

W2. Another 2×2. Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}.$$

(a) Find A^{-1} by reducing $[A \mid I]$.
(b) **Check:** compute $A^{-1}A$ (should be I).
(c) Solve $Ax = b$ for $b = (5, 13)^T$ using $x = A^{-1}b$, then verify $Ax = b$.

W3. When the inverse does **not** exist. Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

(a) Try to reduce $[A \mid I]$ to $[I \mid ?]$. What goes wrong?
(b) Find a nonzero vector in $N(A)$. If $Ax = b$ has one solution, how many does it have?

W3b. General 2×2 formula. For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

row-reducing $[A \mid I]$ (when it works) always produces

$$A^{-1} = \frac{1}{\qquad\qquad} \begin{bmatrix} \qquad & \qquad \\ \qquad & \qquad \end{bmatrix}.$$

The scalar in the denominator is $ad - bc$ (the **determinant** of A — more Friday). If $ad - bc = 0$, there is no inverse (same failure as W3).

Quick check: apply the formula to W1 and W2 and match what you already found.

Column space, rank, and synthesis

W4. Column space. Let

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Describe $C(A)$. Is $b = (1, 2, 3)^T$ in $C(A)$? Is $b' = (1, 3, 0)^T$?

W5. Rank from RREF. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \text{rref}(A) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

What is $\text{rank}(A)$? Give a basis vector for $N(A)$.

W6. Synthesis. For an $m \times n$ matrix, match: existence of a solution; uniqueness (when solvable); rank; when an inverse can exist.

Your turn

Y1. Find A^{-1} for $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ by reducing $[A \mid I]$. Check by computing $A^{-1}A$.

Y2. For $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, give one nonzero vector in $N(A)$ and explain what it means for uniqueness.

Y3. For $A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}$, is $b = (2, 4, -1)^T$ in $C(A)$? Is $b' = (2, 5, -1)^T$?

Y4. An $A \in \mathbb{R}^{3\times 2}$ has rank 2. Is $N(A) = \{0\}$? Does every $b \in \mathbb{R}^3$ have a solution? If a solution exists, is it unique? Is A invertible?

Answers (Your turn)

Y1. Row-reduce $[A \mid I]$ to get

$$A^{-1} = \frac{1}{6}\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}.$$

Check: $A^{-1}A = I$.

Y2. $(2, -1)^T$ (or any nonzero multiple). Any consistent system has infinitely many solutions: add $t(2, -1)^T$.

Y3. Yes — b with $x = (2, -1)^T$. No — the second entry must equal twice the first.

Y4. Yes ($N(A) = \{0\}$); no (not every $b \in \mathbb{R}^3$); yes (unique when solvable); **no**, A is not square, so not invertible.